

Rhombix Technologies

DevOps Internship — Task Submission

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Task Selected	Project 1: Container Orchestration with Kubernetes
Organization	Rhombix Technologies

1. Task Overview

This report documents the completion of Project 1 — Container Orchestration with Kubernetes — assigned as part of the Rhombix Technologies DevOps internship. The objective of this task was to gain hands-on experience with Kubernetes (K8s), the most widely used container orchestration platform.

The task involved enabling Kubernetes on Docker Desktop, deploying a containerized application, managing pods, deployments, and services, scaling the application up and down, and applying configuration using declarative YAML files.

Key Learning Objectives

- Enabling and verifying a local Kubernetes cluster using Docker Desktop (kind provisioning).
- Verifying cluster and node status using `kubectl cluster-info` and `kubectl get nodes`.
- Creating a deployment and verifying pods using `kubectl create deployment` and `kubectl get pods`.
- Exposing a deployment as a service using `kubectl expose` and verifying it with `kubectl get services`.
- Accessing the deployed application in the browser via port-forwarding.
- Scaling a deployment up and down using `kubectl scale deployment --replicas`.
- Writing declarative Kubernetes manifests (`deployment.yaml`, `service.yaml`) and applying them with `kubectl apply -f`.
- Verifying all cluster resources using `kubectl get all` and Docker Desktop's Kubernetes dashboard.

2. Task Execution — Screenshots

Below are the screenshots captured while performing the assigned Kubernetes orchestration task using Docker Desktop's built-in Kubernetes cluster.

2.1 Kubernetes Enabled in Docker Desktop

Kubernetes was enabled from Docker Desktop settings, using the kind cluster provisioning method with a single node running Kubernetes v1.36.1.

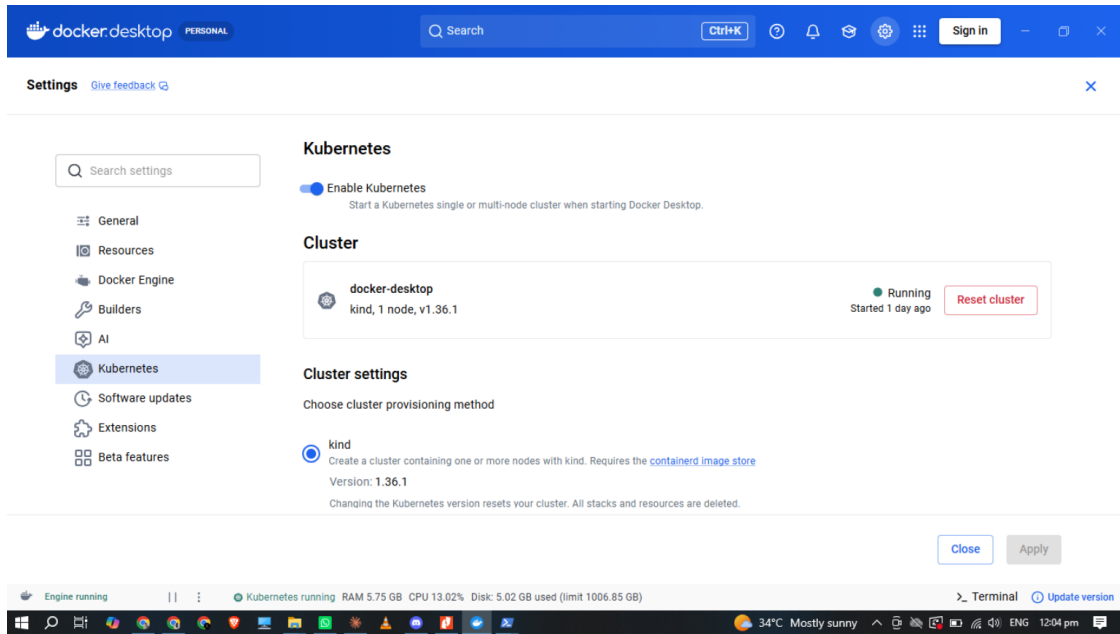


Fig 1: Kubernetes enabled in Docker Desktop settings — kind cluster, 1 node, running

2.2 Cluster, Node, Deployment, Pod and Service Verification

The cluster status was verified using `kubectl cluster-info` and `kubectl get nodes`. A deployment (rhombix-k8s) was created from the nginx image, its pods were verified as Running, and the deployment was exposed as a LoadBalancer service. Port-forwarding was also configured to access the service locally.

```

PS C:\Users\IBKQ> kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:49987
CoreDNS is running at https://127.0.0.1:49987/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
PS C:\Users\IBKQ> kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
desktop-control-plane Ready   control-plane   14m   v1.36.1
PS C:\Users\IBKQ> kubectl create deployment rhombix-k8s --image=nginx
deployment.apps/rhombix-k8s created
PS C:\Users\IBKQ> kubectl get deployments
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
rhombix-k8s 0/1      1             0           15s
PS C:\Users\IBKQ> kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
rhombix-k8s-5c559f95dd-xqwhs 1/1     Running   0           3m53s
PS C:\Users\IBKQ> kubectl expose deployment rhombix-k8s --type=LoadBalancer --port=80
service/rhombix-k8s exposed
PS C:\Users\IBKQ> kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes          ClusterIP   10.96.0.1     <none>          443/TCP    23h
rhombix-k8s         LoadBalancer 10.96.114.70  172.18.0.5     80:32187/TCP 11s
PS C:\Users\IBKQ> kubectl port-forward service/rhombix-k8s 8082:80
Forwarding from 127.0.0.1:8082 -> 80
Forwarding from [::1]:8082 -> 80
Handling connection for 8082
Handling connection for 8082

```

Fig 2: Verifying cluster info, nodes, creating deployment, pods, exposing service, and port-forwarding

2.3 Application Running in Browser

The deployed application was accessed through the browser at localhost:8082 via port-forwarding, confirming that the nginx application was successfully running inside the Kubernetes pod.

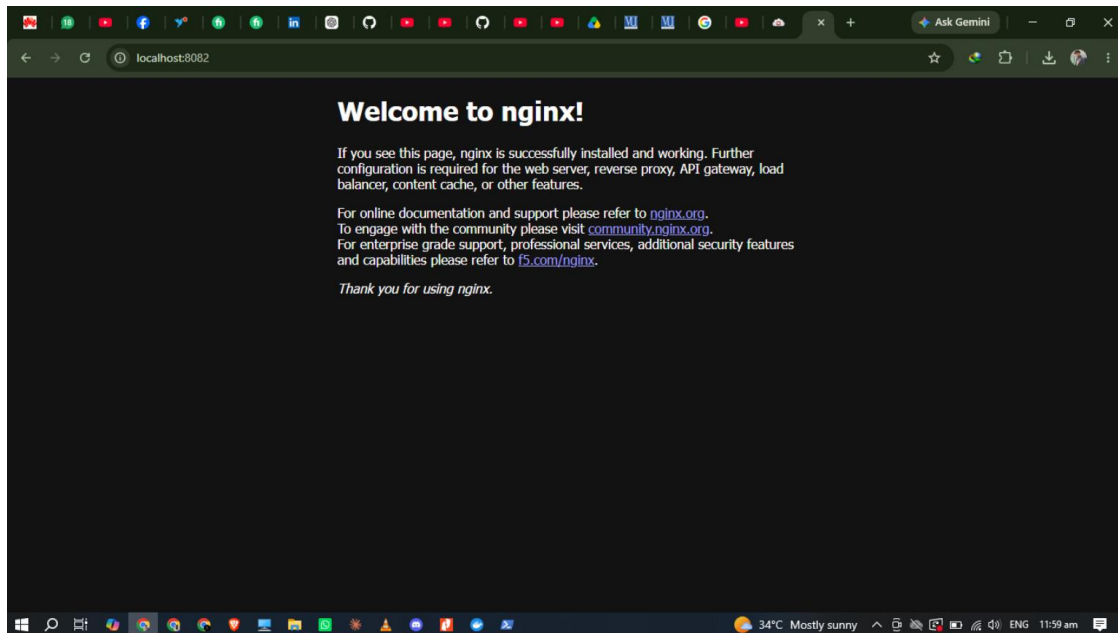


Fig 3: Default nginx welcome page accessed via localhost:8082, confirming the pod is serving traffic

2.4 Scaling the Deployment and Final Verification

The deployment was scaled from 1 to 3 replicas using `kubectl scale deployment --replicas=3`, and the new pods were verified as Running. It was then scaled back down to 1 replica. Finally, `kubectl get all`, `kubectl get deployment`, and `kubectl get service` were used to verify the complete state of the cluster resources.

```
Windows PowerShell
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PS C:\Users\IBKQ> kubectl scale deployment rhombix-k8s --replicas=3
deployment.apps/rhombix-k8s scaled
PS C:\Users\IBKQ> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
rhombix-k8s-5c559f95dd-78rcq        1/1     Running   0           10s
rhombix-k8s-5c559f95dd-fj5qk        1/1     Running   0           10s
rhombix-k8s-5c559f95dd-xqwhs        1/1     Running   0           28m
PS C:\Users\IBKQ> kubectl scale deployment rhombix-k8s --replicas=1
deployment.apps/rhombix-k8s scaled
PS C:\Users\IBKQ> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
rhombix-k8s-5c559f95dd-xqwhs        1/1     Running   0           31m
PS C:\Users\IBKQ> kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/rhombix-k8s-5c559f95dd-xqwhs    1/1     Running   0           34m

NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
service/kubernetes  ClusterIP   10.96.0.1    <none>         443/TCP          24h
service/rhombix-k8s LoadBalancer 10.96.114.70 172.18.0.5    80:32187/TCP     29m

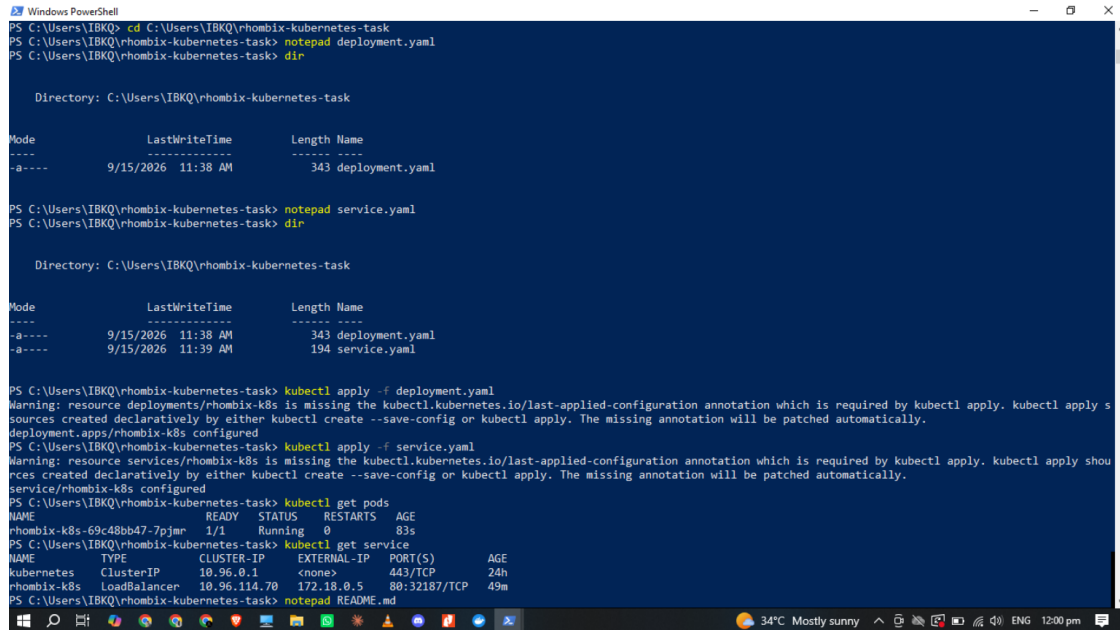
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/rhombix-k8s  1/1     1            1           34m

NAME                DESIRED   CURRENT   READY   AGE
replicaset.apps/rhombix-k8s-5c559f95dd  1         1         1       34m
PS C:\Users\IBKQ> kubectl get deployment
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
rhombix-k8s  1/1     1            1           36m
PS C:\Users\IBKQ> kubectl get service
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1    <none>         443/TCP          24h
rhombix-k8s         LoadBalancer 10.96.114.70 172.18.0.5    80:32187/TCP     31m
PS C:\Users\IBKQ> kubectl get service
```

Fig 4: Scaling deployment from 1 to 3 replicas, back to 1, and verifying all cluster resources with `kubectl get all`

2.5 Kubernetes YAML Files and Apply Verification

Declarative Kubernetes manifests (deployment.yaml and service.yaml) were created and applied using `kubectl apply -f`, demonstrating Infrastructure-as-Code style management of Kubernetes resources. The pods and services were re-verified after applying the YAML configuration.



```
PS C:\Users\IBKQ> cd C:\Users\IBKQ\rhombix-kubernetes-task
PS C:\Users\IBKQ\rhombix-kubernetes-task> notepad deployment.yaml
PS C:\Users\IBKQ\rhombix-kubernetes-task> dir

Directory: C:\Users\IBKQ\rhombix-kubernetes-task

Mode                LastWriteTime         Length Name
----                -
-a-----          9/15/2026 11:38 AM             343 deployment.yaml

PS C:\Users\IBKQ\rhombix-kubernetes-task> notepad service.yaml
PS C:\Users\IBKQ\rhombix-kubernetes-task> dir

Directory: C:\Users\IBKQ\rhombix-kubernetes-task

Mode                LastWriteTime         Length Name
----                -
-a-----          9/15/2026 11:38 AM             343 deployment.yaml
-a-----          9/15/2026 11:39 AM             194 service.yaml

PS C:\Users\IBKQ\rhombix-kubernetes-task> kubectl apply -f deployment.yaml
Warning: resource deployments/rhombix-k8s is missing the kubect1.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply s
sources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
deployment.apps/rhombix-k8s configured
PS C:\Users\IBKQ\rhombix-kubernetes-task> kubectl apply -f service.yaml
Warning: resource services/rhombix-k8s is missing the kubect1.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply shou
rces created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
service/rhombix-k8s configured
PS C:\Users\IBKQ\rhombix-kubernetes-task> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
rhombix-k8s-69c48bb47-7pjmnr        1/1     Running   0           83s
PS C:\Users\IBKQ\rhombix-kubernetes-task> kubectl get service
NAME                                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes                         ClusterIP   10.96.0.1     <none>         443/TCP     24h
rhombix-k8s                        LoadBalancer 10.96.114.70  172.18.0.5    80:32187/TCP 49m
PS C:\Users\IBKQ\rhombix-kubernetes-task> notepad README.md
```

Fig 5: Creating and applying deployment.yaml and service.yaml, then verifying pods and services

2.6 Docker Desktop Kubernetes Resources

The Kubernetes dashboard within Docker Desktop was used to visually confirm the active cluster, deployment status, and running pods, providing a GUI-based verification alongside the `kubectl` command-line outputs.

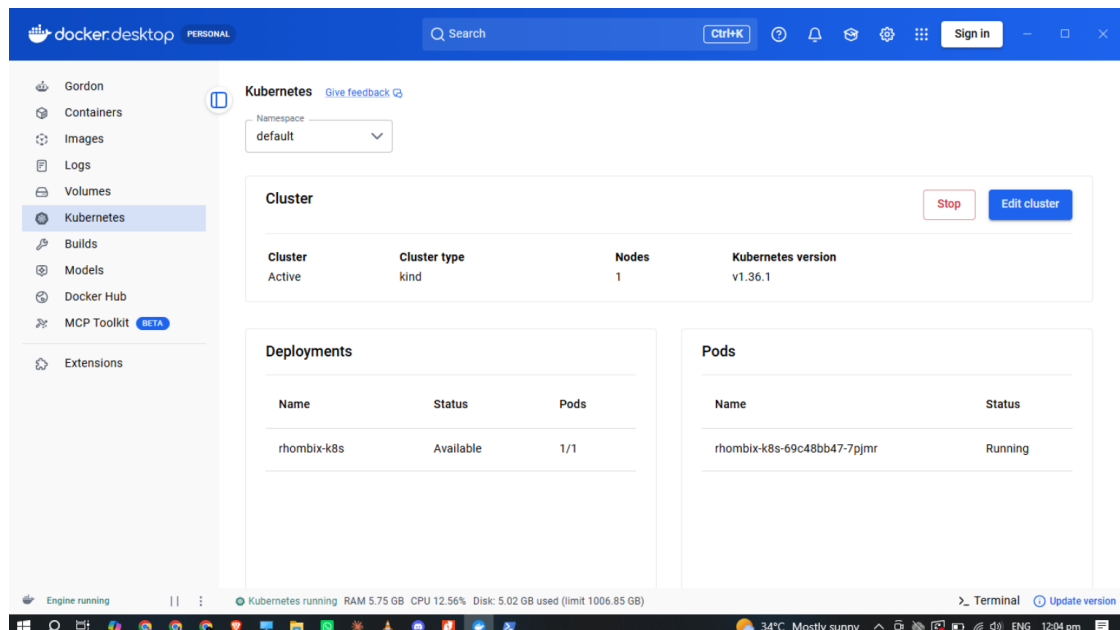


Fig 6: Docker Desktop Kubernetes dashboard showing active cluster, deployment, and running pod

3. Conclusion

Through this task, hands-on experience was gained in container orchestration using Kubernetes. Enabling a local cluster, creating and exposing deployments, scaling applications, and managing resources declaratively through YAML manifests provided a strong foundation in Kubernetes — a core skill for deploying and managing containerized applications in real-world DevOps environments.

This task has been completed as part of the Rhombix Technologies DevOps Internship Program and is being submitted in accordance with the internship submission guidelines.